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Course

Programming
Fundamentals Lab

Assignment:

Theory (2)

Submitted to:

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COMSAT's University Vehari.

Question No 1

Write Pseudocode

"Rock, Paper, scissor" Logic.

Also mention reasons
the choose of specific
structure (nested if-else or
switch case).

Code:

Start

Input player1choice

Input player2choice

If player1choice == player2choice Then
 print "It's a tie"

Else

If player1choice == 'R' Then

If player2 choice == 'S' Then

Print "player 2 wins"

Else

Print "player 2 wins"

ENDIF

Else if player1choice == 'P' Then

```
If player2choice == 'R' Then  
    Print "player1 wins"  
ENDIF
```

```
Else if player1choice == 'S' Then  
    If player2choice == 'P' Then  
        Print "player 1 wins"  
    Else  
        Print "player 2 wins"  
    ENDIF  
ENDIF  
ENDIF  
END
```

Why nested if-else structure:

→ 9 combinations:

There are 9 possible outcomes.

→ Dependent decision:

Result depend on first user's choice then on computer, requiring conditions inside conditions.

→ Better organization of logic:

Nested-if make program structure easier for complex conditions.

→ Avoid writing all combination separately

Question No 2

Pseudocode for Toll Booth Calculator (Switch Case)

Start

Input vehicleType // "Bus", "car", "Truck"

Switch vehicleType

Case "Car":

Print "Toll Tax = 70 Rupees"

Break

Case "Bus.":

Print "Toll Tax = 150 Rupees"

Break

case "Truck"

Print "Toll Tax = 400 Rupees"

Default;

Print "Invalid vehicle Type"

END switch

END

Explanation: (Switch Case)

- Switch is best when one variable has multiple fixed values.
- Cleaner and simple than if-else statements.
- Easy to expand with new vehicle type.

Question No 3

Pseudocode (Leap year validator)

Start

Declare year

Print "Enter year"

Input year

IF (year % 400 == 0)

Print "Leap year"

Else if (year % 100 == 0)

Print "Not a leap year"

Else if ($\text{Year} \% 4 == 0$)

Print "Leap Year"

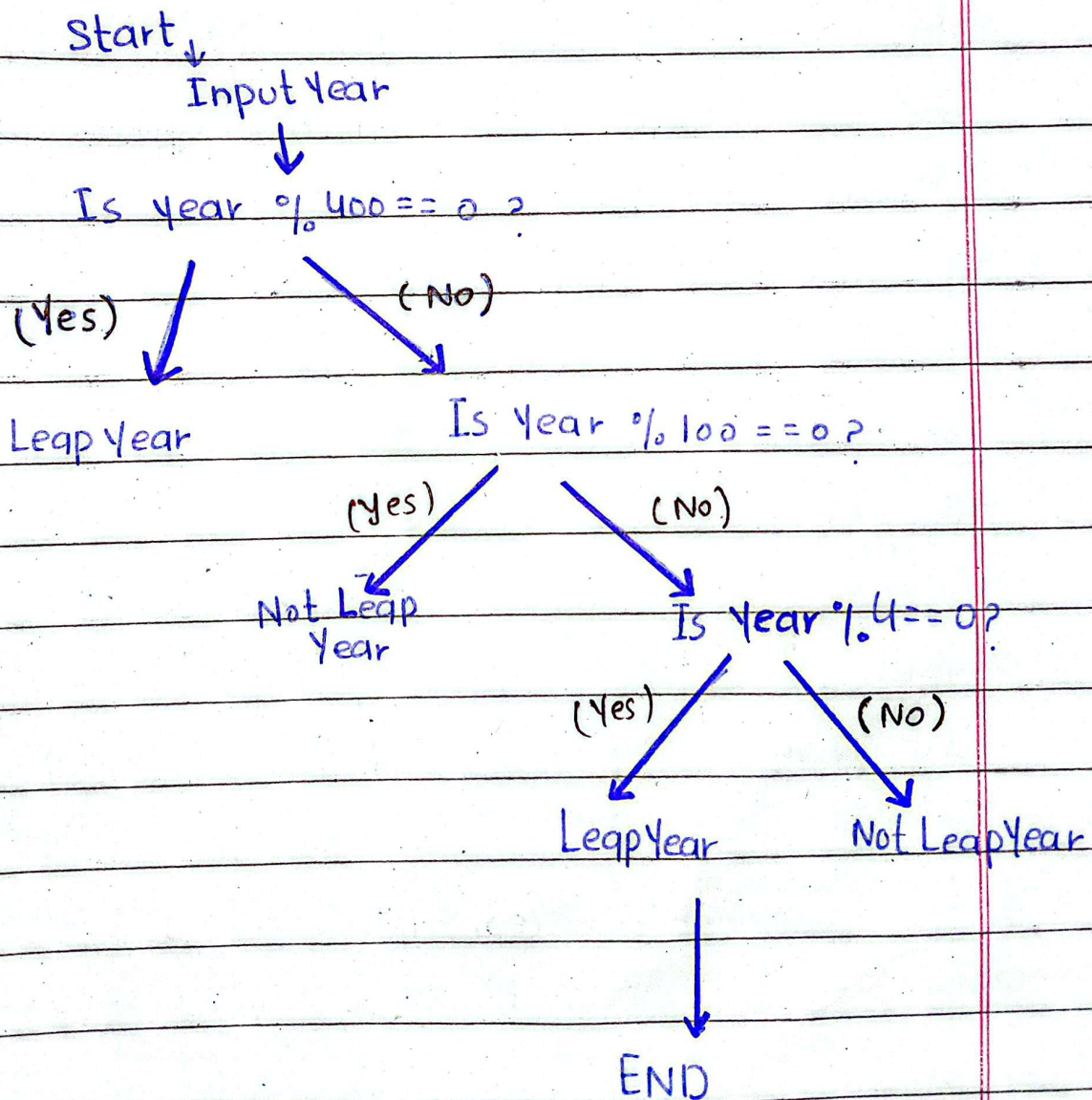
Else

Print "Not a Leap Year"

ENDIF

END

Logical Flow Chart:



Question No 4

Write Java code:

Credit Limit Calculator

```
import java.util.Scanner;

public class CreditLimit {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter Account number");
        int accountNumber = sc.nextInt();
        System.out.println("Enter Monthly Beginning Balance");
        int balance = sc.nextInt();
        System.out.println("Enter Total charges");
        int charges = sc.nextInt();
        System.out.println("Enter Total credits");
        int credits = sc.nextInt();
        System.out.println("Enter credit limit");
        int creditLimit = sc.nextInt();
        int newBalance = balance + charges - credits;

        if (newBalance > creditLimit)
        {
            System.out.println("Credit Limit exceeds");
        }
        else
        {
            System.out.println("New Balance" + newBalance);
        }
    }
}
```

```
}  
}
```

Question No 5

Develop Java Application

Sales Commision Calculator

```
import java.util.Scanner;  
  
public class SalesCommission{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);  
  
        double totalSales = 0;  
        double itemPrice = 1;  
  
        System.out.println("Enter item prices. Type 0  
                            when you done");  
  
        while (itemPrice != 0) {  
            System.out.println("Item price");  
            itemPrice = sc.nextDouble();  
            totalSales = totalSales + itemPrice;  
        }  
    }  
}
```

```

double bonus = totalSales * 0.09;
double finalPay = 200 + bonus;
System.out.println("Total sales" + totalSales);
System.out.println("Your total pay for this week is: $" +
                    finalPay);
    }
}

```

Question No 6

Java Application for three employees. Salary Calculator

```

import java.util.Scanner;

public class salaryCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        for (int i = 1; i <= 3; i++) {
            System.out.println("Enter hours worked for employee");
            double hours = input.nextDouble();
            System.out.println("Enter hourly rate");
            double rate = input.nextDouble();
            double grossPay;

```

```

if( hours <= 40 )
{
    grossPay = hours * rate ;
}
else
{
    grossPay = (40 * rate) + ((hours - 40) * (
        rate * 1.5 ));
}
}
}
}
System.out.println("Gross Pay for employee " +
    GrossPay);
}
}
}

```

Question No 7
(Find largest Number) from 10 numbers
 Pseudocode also Java Code.

Pseudocode:

Start

initialize counter to 0

initialize largest number to very small number
 (or first input)

While counter < 10

input number

if number > largest, then set

largest = number
Increment counter.
Print largest.
End.

Java Code

```
import java.util.Scanner;

public class FindLargest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int counter = 0;
        int largest = Integer.MIN_VALUE;
        while (counter < 10) {
            System.out.println("Enter integer" +
                               (counter + 1));
            int number = sc.nextInt();
            if (number > largest)
            {
                largest = number;
            }
            counter++;
        }
        System.out.println("The largest number" +
                           largestNumber);
    }
}
```

Question No 8 (Dangling Else problem)

Modified version for provided code.

(a)

```
if (y==8){  
    if (x==5){  
        System.out.println("@@@@@")  
    }  
}  
  
System.out.println("$$$$$")  
System.out.println("$$$$$")
```

(b)

```
if (y==8){  
    if (x==5){  
        System.out.println("@@@@@");  
    }  
}
```

(c)

```
if (y==8){  
    if (x==5){  
        System.out.println("@@@@@");  
    }  
}
```

(d)

```
if (y==8){  
    if (x==5){  
        System.out.println("@@@@@");  
    }  
    else  
    {  
        System.out.println("#####");  
        System.out.println("$$$$$");  
        System.out.println("88888");  
    }  
}
```

Question No 9 (Calculate product of odd Integers)

```
Public class oddProduct{  
    public static void main(String[] args){  
        int product=1;  
        for(int i=1; i<=15; i+=2)  
        {  
            product *= i;  
        }  
        System.out.println("Product of odd integers  
        1 to 15 "+ Product);  
    }  
}
```

Question No 10

Write Program:
(Multiplication Table)

```
import java.util.Scanner;
public class MultiTable {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a number to print
                           its table");
        int number = input.nextInt();
        for (int i = 1; i <= 10; i++)
        {
            System.out.println (number + i + " = " + (number * i));
        }
    }
}
```